

1. Select all of the expressions that are equivalent to the value of $2y - 4x$ when $x = 5$ and $y = -1$.

- ☒ -22
- ☐ -18
- ☒ $(-2) - 20$
- ☒ $(-2) + -20$
- ☐ $2(-1) + 4(5)$
- ☒ $2(-1) - 4(5)$

2. A candle weighs 300 grams (g). Each hour the candle is lit, it burns 4 g of its wax. Let h be the number of hours the candle burns.

Write an expression that represents the weight of the candle.

Expression	$300 - 4h$
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3. Select all of the phrases that represent the algebraic expression shown.

$$y - 14$$

- ☒ fourteen less than y
- ☐ y less than fourteen
- ☒ the sum of y and -14
- ☐ y subtracted from -14
- ☒ the difference between y and 14

4. Select all of the expressions that are equivalent to the $3x + 9(2y - x)$.

- ☐ $2x + 18y$
- ☒ $18y - 6x$
- ☐ $18y - x + 3x$
- ☒ $9(2y - x) + 3x$
- ☐ $-2(3x - 8y) - 2y$

5. Evaluate the expression. If necessary, use the space provided to show your work.

Expression	Work
$8h \div 10^3$ when $h = -500$	
Value	
-4	

6. Select all of the expressions that are equivalent to $8h + 9k - 6h + 24k$.

- ☒ $2h + 33k$
☐ $14h + 33k$
☒ $8(3k + h) + 3(3k - 2h)$
☒ $(8h - 6h) + (9k + 24k)$
☐ $3(8k + 3k) - 2(-2h)$

7. Write an expression to represent the phrase " n pounds of sugar equally separated into 8 cups".

Expression	$\frac{n}{8}$
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8. Which value of x is the solution to $-6x + 22 = 8(1 - x)$?

A. -7

B. -3

C. 1

D. 2

9. Select all of the values of x that are solutions to $17 - 7x > 3(10x - 4) + 61$.

☒ -4

☒ -2

☒ -1

☐ 0

☐ 2

☐ 5

10. Find the sum or difference of each expression using the fewest terms possible.

a. $(x + 9) + (2x + 3)$

$$3x + 12$$

b. $(11k + 15) - (7k + 8)$

$$4k + 7$$

c. $14.8 - (6.3 - 2.9r)$

$$2.9r + 8.5$$

11. What is the result when $\frac{7}{8}w + \frac{5}{6}$ is subtracted from $\frac{14}{4}w - \frac{23}{6}$?

$$\frac{21}{8}w - \frac{28}{6}$$

12. Use the given expressions to answer each.

$$14(2m + 5) - 2 \text{ and } 28m + 3$$

- a. Choose a value for m to determine if the expressions are equivalent. Show your work.

Answers will vary. Use of $m = 0$ will show equivalence and is the only value that should not be used.

- b. Use properties of operations to determine if the expressions are equivalent. Show your work.

$$28m + 70 - 2$$

$$28m + 68$$

- c. Complete the statement.

The expressions

☐ are

☒ are not

equivalent.